Lecture 14 15

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1 Counting Principles

1.1 The Sum Rule

If there are m ways to choose element a and n ways to choose element b (independent of a), then there are $m + n$ ways to choose a or b.

- **Set Theory Notation**: $|A \cup B| = |A| + |B|$ (if sets are disjoint).

1.1.1 Example: Path Counting

Consider a graph with nodes A, B, C, D.

- **Path 1**: A to C via B.
  - A to B: 3 paths.
  - B to C: 4 paths.
  - Total ways via B: $3 \times 4 = 12$.
- **Path 2**: A to C via D.
  - A to D: 2 paths.
  - D to C: 3 paths.
  - Total ways via D: $2 \times 3 = 6$.

Problem: How many ways from A to C?

$$3 \times 4 + 2 \times 3 = 12 + 6 = 18$$

1.2 The Product Rule

If there are m ways to choose element a and n ways to choose element b, then there are $m \times n$ ways to choose the pair (a, b).

- **Set Theory Notation**: $|A \times B| = |A| \cdot |B|$.

1.2.1 Problem: 6-digit Numbers

How many 6-digit numbers are there?

- The first digit cannot be 0 (options: 1-9 $\rightarrow$ 9 ways).
- The remaining 5 digits can be 0-9 (10 ways each).

$$9 \cdot 10^5$$

1.2.2 Problem: Even Digits Constraint

How many 6-digit numbers contain at least one even digit?

- **Strategy**: Use the complement (Total numbers - Numbers with only odd digits).
- Odd digits: $\{1, 3, 5, 7, 9\}$ (5 options).
- Numbers with only odd digits: 5^6 (Assuming 6 positions filled with 5 options each).

$$9 \cdot 10^5 - 5^6$$

- **Note**: The term 5^6 assumes the first digit (being odd) is valid. Since 0 is even, it is not in the set of odd digits, so the first digit restriction (non-zero) is automatically satisfied by choosing from $\{1, 3, 5, 7, 9\}$.

1.2.3 Problem: Divisibility by 5

How many 6-digit numbers are divisible by 5?

- Last digit must be 0 or 5 (2 options).
- First digit: 9 options.
- Middle digits: 10 options.

$$9 \cdot 10^4 \cdot 2 = 18 \cdot 10^4$$

2 Permutations and Arrangements

2.1 Permutations

Let S be a set with n elements. The number of arrangements of its elements in order is $n!$.

$$n! = 1 \cdot 2 \cdot \cdots \cdot n$$

- **Example**: $S = \{a, b, c\}$.
  - Permutations: $abc, acb, bca, bac, cab, cba$.
  - Total: $3! = 6$.

2.2 Ordered Selections (k-Permutations)

The number of ordered selections of m elements from a set S of size n.

$$P(n, m) = n \cdot (n-1) \cdot \cdots \cdot (n-m+1) = \frac{n!}{(n-m)!}$$

2.2.1 Example: Selecting Officers

From 100 students, choose a President, VP, and WP (positions are distinct/ordered).

$$100 \cdot 99 \cdot 98$$

2.3 Unordered Selections (Combinations)

The number of unordered selections of m elements from a set S of size n.

$$\binom{n}{m} = \frac{P(n, m)}{m!} = \frac{n!}{m!(n-m)!}$$

- **Notation**: $C_n^m = C(n, m) = \binom{n}{m}$.
- This represents the number of subsets of cardinality m.

2.3.1 Example: Selecting Interviewees

From 100 students, choose 3 for interviews (no distinct roles/unordered).

$$\binom{100}{3} = \frac{100 \cdot 99 \cdot 98}{6}$$

3 Binomial Coefficients and Identities

3.1 Basic Properties

- $\binom{n}{0} = 1$
- $\binom{n}{1} = n$
- $\binom{n}{n} = 1$
- $\binom{n}{2} = \frac{n(n-1)}{2}$
- **Symmetry**: For $0 \leq m \leq n$, $\binom{n}{m} = \binom{n}{n-m}$.

3.2 Applied Problems

3.2.1 Handshake Problem

If n people at a party all shake hands exactly once, how many handshakes occurred?

$$\binom{n}{2} = \frac{n(n-1)}{2}$$

3.2.2 Committee Selection

There are 2 Math professors and 8 CS professors. We need a committee of 6, with at least 1 Math professor.

- **Method 1 (Direct Sum)**:

$$(1 \text{ Math} \wedge 5 \text{ CS}) + (2 \text{ Math} \wedge 4 \text{ CS})$$

$$\binom{2}{1}\binom{8}{5} + \binom{2}{2}\binom{8}{4}$$

- **Method 2 (Complement)**: Total committees - Committees with no Math professors.

$$\binom{10}{6} - \binom{8}{6}$$

3.2.3 Candy Exchange

Alice has 6 different chocolates. Bob has 8 different gummy bears. They want to exchange 5 candies.

- Alice chooses 5 to give: $\binom{6}{5}$.
- Bob chooses 5 to give: $\binom{8}{5}$.
- Total ways: $\binom{6}{5} \cdot \binom{8}{5}$.

3.3 The Binomial Theorem

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

3.3.1 Examples

- $(x+y)^2 = x^2 + 2xy + y^2$
- $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$

3.3.2 Coefficient Problem

What is the coefficient of x^5y^8 in $(x+y)^{13}$?

- Here $n = 13$. We need y^8, so $k = 8$.
- Solution: $\binom{13}{8}$ (which is equal to $\binom{13}{5}$).

3.4 Pascal's Identity

For $0 \leq m \leq n$:

$$\binom{n}{m} = \binom{n-1}{m-1} + \binom{n-1}{m}$$

3.4.1 Proofs

1. **Algebraic**: Expand the factorials $\frac{(n-1)!}{(m-1)!(n-m)!} + \ldots$ and simplify.
2. **Combinatorial**: Let $S = \{a_1, \ldots, a_n\}$. To choose a subset of size m:
   - **Case 1**: Subset contains a_n. We must choose remaining $m-1$ elements from the other $n-1$. $\to \binom{n-1}{m-1}$.
   - **Case 2**: Subset does not contain a_n. We must choose all m elements from the other $n-1$. $\to \binom{n-1}{m}$.

3.5 Sum of Binomial Coefficients

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

3.5.1 Proofs

1. **Combinatorial**: The LHS counts subsets of size 0, 1, ..., n. The total number of subsets of a set with size n is 2^n.

2. **Algebraic**: Use the Binomial Theorem with $x = 1, y = 1$.

$$(1+1)^n = \sum_{k=0}^{n} \binom{n}{k} 1^{n-k} 1^k = 2^n$$

4 Integer Solutions (Stars and Bars)

Problem: How many solutions for $x + y + z = 10$ exist in non-negative integers?

4.0.1 Method

We can represent the number 10 as ten ones (stars) and use dividers (bars) to separate variables.

- Example Solution: $x = 1, y = 0, z = 9$.
- Representation: 1||111111111
- Total items to arrange: 10 stars + 2 bars = 12 positions.
- We choose positions for the 2 bars.

$$\binom{12}{2}$$

5 The Inclusion-Exclusion Principle

5.1 Two Sets

$$|A \cup B| = |A| + |B| - |A \cap B|$$

5.1.1 Problem: Student Programs

Total students: 160.

- 80 chose Math (M).
- 70 chose Programming (P).
- 40 chose both ($M \cap P$).

How many chose neither?

- $|M \cup P| = 80 + 70 - 40 = 110$.
- Neither $= 160 - 110 = 50$.

Note: The handwritten source contains a calculation error or typo ($160 - 150 = 10$), likely miscalculating the union. The standard Inclusion-Exclusion calculation is provided above based on the given subset values.

5.2 Three Sets

$$|A \cup B \cup C| = |A| + |B| + |C| - (|A \cap B| + |A \cap C| + |B \cap C|) + |A \cap B \cap C|$$

5.2.1 Problem: Divisibility

How many numbers $n \leq 1000$ are divisible by 7 or 11? Define sets:

- $A = \{n \leq 1000 : 7|n\}$
- $B = \{n \leq 1000 : 11|n\}$

Calculate cardinalities:

- $|A| = \lfloor \frac{1000}{7} \rfloor = 142$
- $|B| = \lfloor \frac{1000}{11} \rfloor = 90$
- $|A \cap B| = \lfloor \frac{1000}{77} \rfloor = 12$

Calculate Union:

$$|A \cup B| = 142 + 90 - 12 = 220$$

Note: The handwritten notes show "80" for $|B|$ in the final summation line, which is inconsistent with $\lfloor 1000/11 \rfloor = 90$. The correct value (90) is used here to match the final result of 220.